

EDUCATION

Bachelors of Science in **Professional Physics**. Montana State University, 2018. **3.95 GPA**

WORK EXPERIENCE

Emotive

Senior Software Engineer

November 2021 - **Present**

- Transitioned from monolith to microservice architecture to ensure scalability while **maintaining a cyclomatic complexity of 5 or lower**. This made it possible for teams to function more independently and increased the org's velocity
- Refactored systems to **reduce query times by 99.9%** by introducing covering indices and lateral joins into our PostgreSQL database. This allowed the database to be downsized which lowered operational costs.
- Worked with finance team to lead a project automating the import + upload + charging of monthly billing spreadsheets. This **prevented the org from doing 100s of hours of manual, repetitive, error-prone tasks**
- Rewrote legacy React components to **remove unnecessary global state** + abstractions allowing for a **75% reduction in total lines of code** for a microfrontend
- Lead the company's "Frontend Guild"** to educate and elevate the organization's frontend engineers. We explored topics like the transition to React 18, when to use various state management solutions, as well as a general best practice consensus.

15Five Inc

Engineering Manager

March 2021 - Oct 2021

- Led a team of 4 engineers tasked with maintaining and **improving the core 15Five Check-in feature** of our app.
- Cultivated a healthy atmosphere by encouraging direct reports to bring their full selves to work through regular 1-on-1's, performance reviews, and role clarity discussions which led to **100% retention of the engineers on my team**.
- Collaborated with designers, product managers, mobile engineers, and other teams** while participating in customer calls to incorporate feedback, build new features, and integrate other teams' work into the Check-in.

Technical Lead

Oct 2020 - Oct 2021

- Developed end to end solutions** for all feature work. We typically began by establishing scope before writing and presenting an implementation plan and tickets. During the development cycle I would **write code and review 100% of the relevant pull requests**.
- Implemented TypeScript adoption at an org level. I worked with several other developers to **enforce static typing on 100% of our React components**. This was part of the effort that allowed our DevOps team to embrace CI/CD and **decrease cycle time by 50% org-wide**.
- Architected and built an extensible framework to simplify app-wide rich text editing. I began by investigating existing usage in order to determine the proper abstractions before **removing 1000s of lines of duplicated code and several abstraction layers** from our codebase.

Frontend Engineer

Oct 2019 - Oct 2020

- Built OKR tracking feature into existing app infrastructure. This feature has **grown into our most adopted feature**. In order to implement this, we had to **integrate custom React hooks and functional components into legacy jQuery** business logic.

Triple Tree LLC

Software Engineer

Mar 2019 - Aug 2019

- Built a veteran transition program with the COMMIT Foundation. **Implemented OAuth2** to ensure data security and developed a video streaming and recommendation engine by **transforming client designs into React + Redux components**.
- Researched, Designed, & Implemented app wide **email automation logic**, canary releases on AWS, a **video recommendation engine** built in NodeJS

The Business Garage

Junior Software Engineer

Feb 2018 - Mar 2019

- Built the backend and **managed deployments for a web app with 50K users**. We experienced viral growth and **implemented redis caching to reduce load times by 60%**. Prior to the growth, I ensured scalability by developing distributed load test infrastructure through AWS.
- Researched, Designed, & Implemented **push notifications** through a redis queue, **content caching** to reduce db stress, **distributed load testing** on AWS

LIGO Scientific Collaboration— Researcher

2013, 2016 - 2018

- Wrote applications + received grant funding to develop deep neural networks and **model neutron stars** to test Einstein's theory of gravity.

OTHER PROJECTS, EXPERIENCE & SKILLS

Skills

- Used daily at present—Docker, JavaScript, TypeScript, React (with concurrency!), Python, Django, Django Filter, Django REST Framework, Django Channels, Django Celery Beat, Django Debug Toolbar, Git, Arch Linux, Sass, Vim, PostgreSQL
- Used daily in the past—AWS, AdonisJs, Bash, Bokeh, D3, Docker, Express.js, FCM, Hapijs, Heroku, Jupyter, Keras, Kubernetes, MariaDB, Mathematica, Matplotlib, NLTK, Node.js, Numpy, Pandas, Redis, Redux, RethinkDB, SKLearn, SQLite, SciPy, Seaborn, Selenium, Tensorflow, CouchDB

Open source contributions— github.com/jpribyl & johnpribyl.com

- Wrote, published, and maintain `react-hook-dragula` and `react-hook-mathjax` which have received around 20 stars and 300 downloads per week. I have also **contributed to React, Django**, `vim-test`, and `Freqtrade`.
- Forked, fixed a deprecation warning, and released `action-docker-layer-caching` when unable to reach the original maintainer.

Teaching & Coaching

- A combined 8 years of experience coaching soccer and tutoring math and physics up to a college level and have worked with ages ranging from 5 to 50 years and **subjects ranging from arithmetic to quantum mechanics**.

Grand Canyon Rafting

- Successfully led a 3 week, unguided raft trip down 280 river miles through the canyon. I planned all the logistics for 16 people.